

Reachable sets in a thin-layer limit

A reduced theorem, with proof, for a model whose passes concentrate

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Abstract

We study the reachable set of a family of control systems in which a vanishing parameter ε concentrates the forcing: the vehicle crosses a layer of width ε in which a force of size ε^{-1} acts for a time of order ε , and so changes the state by a fixed amount. The limit of such a family forgets these passes, since each has vanishing duration, and yet it is in the passes that the trajectories of the family are told apart. This document isolates the difficulty in one model and proves one theorem about it: the reachable set of the family is contained in that of a limit system, formed by the arcs between passes and by one jump per pass, thickened by a ball whose radius is explicit and vanishes with ε . The limit system is the *skeleton* and its *bubbles*; the ball is the *residual*. The inclusion holds uniformly over the controls, over the initial set, and over a bounded family of admissible vector fields, which is the sense in which an outer bound must hold if it is to be a certificate. The proof is elementary throughout: a dissipation identity supplies a budget, the budget bounds both the number of passes and the time spent in the layer, one change of variable makes the inner problem independent of ε , and Gronwall's inequality assembles the pieces. [Section 7](#) states how hypersonic entry instantiates the model, which hypothesizes it satisfies and which it does not. Only the theorem of [Section 4](#) is claimed as proved.

1 The question, reduced

The object we want enclosed is a reachable set: every state a control system can be driven to, under every admissible control and every admissible model of its forcing. For an atmospheric entry vehicle this is what a safety analysis needs, and two things stand in the way — the dimension of the state, and the provenance of the guarantee, since a bound produced by a numerical solver inherits that solver's accuracy. Our program addresses both, and [Section 7](#) describes it. This document does something narrower and prior to it: it takes the one analytical difficulty on which the program turns, strips it of the application, and proves a theorem about it.

The difficulty is a singular limit. Entry supplies a small parameter, the ratio $\varepsilon = H_s/R_e$ of the atmospheric scale height to the planetary radius. Measured on the planetary length scale, the sensible atmosphere is a layer of thickness ε , and the aerodynamic force inside it is of size ε^{-1} . A vehicle that dips into the layer therefore spends a time of order ε there and leaves with its velocity changed by a fixed amount. Call such a dip a *pass*. As $\varepsilon \rightarrow 0$ a pass becomes instantaneous and keeps its effect, so that in the limit a trajectory is a sequence of arcs joined by jumps — and yet the natural limiting objects do not see the jumps, since an occupation measure weighs an interval of time by its length and a pass has no length in the limit. A reachable set is a support and not an integral, so what has no mass still counts.

The remedy we propose is a decomposition: a low-dimensional *skeleton*, which is what survives the limit; one *bubble* per pass, the pass rescaled to its own width, whose net effect on the surviving coordinates is a jump; and a *residual*, which is what the skeleton and the finitely many bubbles do not capture. The statement we

want is that the reachable set of the family is contained in that of the skeleton-and-bubble system, thickened by the residual. This document asks whether that statement is true, and with what residual, in a model that keeps the concentration and discards everything else.

Remark 1.1 (Why the forcing must be of size ε^{-1}). It is tempting to write the reduced problem as a slow state driven by a fast one, $\dot{z} = g(z) + h(y)$ with $\varepsilon \dot{y} = q(y, z)$ and h bounded. That form does not contain the difficulty. If h is bounded and the excursion in y lasts a time of order ε , its contribution to z is of order ε , so in the limit the excursion leaves no trace and there is no bubble to extract: the limit is the slow flow alone, and ordinary averaging describes it. Concentration requires the forcing to be of size ε^{-1} on a set of width ε , so that its integral is of order one while its support shrinks to a point. The model below is scaled that way, and the scaling is the one the atmosphere supplies rather than one we impose: with altitude measured on the planetary scale, an exponential atmosphere of scale height ε has

$$\int_0^\infty \varepsilon^{-1} e^{-y/\varepsilon} dy = 1 \quad \text{for every } \varepsilon > 0, \quad (1.1)$$

so the aerodynamic impulse available in the layer is of order one, however thin the layer is.

What this document does not do should be said at the start. It does not prove the theorem we want for entry; [Section 7](#) lists what separates the two. It does not use occupation measures: the theorem of [Section 4](#) is a statement about trajectories, proved by elementary means, and [Section 6](#) explains where the measures do enter, namely in certifying the reduced problem that the theorem hands them. And it does not use concentration-compactness. We had expected to need it, and it remains the natural tool for the structure in the general case; but in the model below the number of passes is bounded in advance by a dissipation budget, and a budget is a stronger thing to have than a compactness argument, because it comes with a constant.

2 The model and its hypotheses

Fix an integer $d \geq 1$ and a horizon $T > 0$. For $\varepsilon \in (0, 1]$ the state is $x = (y, v, s)$, with $y \in \mathbb{R}$ an altitude, $v \in \mathbb{R}$ its rate, and $s \in \mathbb{R}^d$ a vector of slow coordinates. Write

$$\rho_\varepsilon(y) = \varepsilon^{-1} e^{-y/\varepsilon} \quad (2.1)$$

for the density of the layer. The system is

$$\dot{y} = v, \quad (2.2)$$

$$\dot{v} = a(s, c, \theta) + \rho_\varepsilon(y) L(s, c, \theta), \quad (2.3)$$

$$\dot{s} = F(s, c, \theta) + \rho_\varepsilon(y) \Gamma(s, c, \theta). \quad (2.4)$$

Here $c(\cdot)$ is the control and $\theta(\cdot)$ is a selection, that is, a choice of admissible vector field from a bounded family; both are measurable functions of time, and the distinction between them is only that we must bound uniformly over both. We abbreviate the two parts of the field as

$$f_0(x, c, \theta) = (v, a, F), \quad b(s, c, \theta) = (0, L, \Gamma),$$

so that [Eqs. \(2.2\) to \(2.4\)](#) read $\dot{x} = f_0(x, c, \theta) + \rho_\varepsilon(y) b(s, c, \theta)$. We call f_0 the *skeleton field*, because it is what remains where the layer is not felt, and b the *layer field*. Four hypotheses are imposed.

Assumption 2.1 (A1: corridor and data). The control set $\mathcal{U} \subset \mathbb{R}^m$, the selection set $\Theta \subset \mathbb{R}^p$ and the slow corridor $S \subset \mathbb{R}^d$ are compact. The functions $a, L: S \times \mathcal{U} \times \Theta \rightarrow \mathbb{R}$ and $F, \Gamma: S \times \mathcal{U} \times \Theta \rightarrow \mathbb{R}^d$ are continuous, bounded by $a_{\max}, L_{\max}, F_{\max}, \Gamma_{\max}$, and Lipschitz in s uniformly in (c, θ) with constants $\kappa_a, \kappa_L, \kappa_F, \kappa_\Gamma$. A trajectory is an absolutely continuous $x: [0, T] \rightarrow \mathbb{R}^2 \times \mathbb{R}^d$ satisfying [Eqs. \(2.2\) to \(2.4\)](#) almost everywhere and remaining in the compact corridor

$$\mathcal{X} = \{ (y, v, s) : |y| \leq y_{\max}, |v| \leq v_{\max}, s \in S \}.$$

Its initial value lies in a set $\mathcal{X}_0 \subseteq \mathcal{X}$ on which $y \geq \ell_1$, with $\ell_1 > 0$ fixed in [Assumption 2.4](#).

Assumption 2.2 (A2: signs). There are $a_{\min} > 0$ and $L_{\min} > 0$ with $a(s, c, \theta) \leq -a_{\min}$ and $L(s, c, \theta) \geq L_{\min}$ for all (s, c, θ) .

The skeleton field accelerates the altitude downward and the layer field upward: gravity pulls and the layer pushes back. Without the second sign a trajectory that enters the layer need never leave it, and without the first it need never enter.

Assumption 2.3 (A3: dissipation budget). There is $\Theta_d \in C^1(\mathbb{R}^d)$ and a constant $c_0 > 0$ with

$$\nabla \Theta_d(s) \cdot F(s, c, \theta) \leq 0, \quad \nabla \Theta_d(s) \cdot \Gamma(s, c, \theta) \leq -c_0,$$

for all (s, c, θ) . We write $\Delta \Theta_d = \max_S \Theta_d - \min_S \Theta_d$ and

$$M = \Delta \Theta_d / c_0. \tag{2.5}$$

This is the one hypothesis the application supplies exactly, and it is the backbone of everything that follows. It says that some function of the slow state decreases along every trajectory, at a rate proportional to the layer density. In entry it is the specific mechanical energy: the lift does no work, so the energy falls at the rate of the drag power, and the fall is strictly proportional to the density. The consequence, drawn in Eq. (5.2) below, is that the time-integrated density along any trajectory is at most M , whatever the control, the selection and the value of ε . A budget of this kind is what replaces a compactness argument: it bounds how many passes there can be, and it does so with a number.

Assumption 2.4 (A4: separated passes). There are $\ell_1 > 0$ and $v_* \in (0, v_{\max}]$ such that along every trajectory:

- (i) $y(0) \geq \ell_1$;
- (ii) $|v(t)| \geq v_*$ whenever $y(t) = \ell_1$;
- (iii) on each connected component of $\{t \in [0, T] : y(t) < \ell_1\}$, the function v vanishes at most once.

We call these components the *passes*.

The first two conditions make the section $y = \ell_1$ transversal, with a quantitative margin: no trajectory grazes it. The third says that a pass is one dip and not a train of them. Together they are what keeps the passes countable and separated, and they are the hypotheses that the application does not satisfy; Section 7 says why, and what the physical phenomenon is that they exclude. We state them as hypotheses rather than hiding them, because the question of what exactly a pass is, and of what rules out an accumulation of passes, is where the general statement is hard. Note that Assumption 2.4 does not assume the passes are finite in number: that is derived, in Eq. (5.3), from the budget.

Remark 2.5 (What is uniform). Every constant named in Assumptions 2.1 to 2.4 is a bound over $S \times \mathcal{U} \times \Theta$. No constant below depends on the particular control $c(\cdot)$, the particular selection $\theta(\cdot)$, or the particular initial point. This is the sense in which the theorem is uniform over the admissible family, and it is the reason the selection θ is carried along at all: an enclosure of an uncertain field is exactly a bounded family of admissible fields, and a bound that held only along a nominal trajectory would certify nothing.

3 The bubble

A pass is short in time but not small in effect, so the right independent variable inside it is not time but the delivered impulse. This one change of variable is what makes the inner problem independent of ε , and it is the only device in the proof that is not completely standard.

Along a trajectory set

$$\mu(t) = \int_{t_0}^t \rho_\varepsilon(y(r)) \, dr, \quad \zeta(t) = e^{-y(t)/\varepsilon} = \varepsilon \rho_\varepsilon(y(t)). \tag{3.1}$$

Then $d\mu/dt = \zeta/\varepsilon > 0$, so μ is a legitimate change of variable, and ζ is a rescaled density, equal to one on the section $y = 0$ and growing below it. Dividing Eqs. (2.2) to (2.4) by $d\mu/dt$ gives, exactly,

$$\frac{d\zeta}{d\mu} = -v, \quad \frac{dv}{d\mu} = L + \frac{\varepsilon}{\zeta} a, \quad \frac{ds}{d\mu} = \Gamma + \frac{\varepsilon}{\zeta} F. \quad (3.2)$$

Two features of Eq. (3.2) carry the proof. The first equation contains no ε at all. And the perturbing terms carry the factor ε/ζ , for which

$$\frac{\varepsilon}{\zeta} d\mu = dt, \quad (3.3)$$

so that their integrated effect is the skeleton drift over the elapsed *time* — which is short, because a pass is short. Setting $\varepsilon = 0$ in Eq. (3.2) leaves a system with no small parameter.

Definition 3.1 (Inner system and pass map). The *inner system* is

$$\frac{d\zeta}{d\mu} = -v, \quad \frac{dv}{d\mu} = L(s, c, \theta), \quad \frac{ds}{d\mu} = \Gamma(s, c, \theta). \quad (3.4)$$

Given $v^- < 0$ and $s^- \in S$, an *inner trajectory* from (v^-, s^-) is a solution of Eq. (3.4) with $(\zeta, v, s)(0) = (0, v^-, s^-)$ and measurable $c(\cdot) \in \mathcal{U}$, $\theta(\cdot) \in \Theta$; its *exit time* is $I = \inf\{\mu > 0 : \zeta(\mu) = 0\}$. The *pass map* is the set-valued map

$$J(v^-, s^-) = \{ (v(I), s(I)) : \text{inner trajectories from } (v^-, s^-) \}.$$

The pass map is the bubble, written as the reachable set of a problem in $2 + d$ coordinates that carries no small parameter. It is the answer to the question of what a pass does: not an impulse chosen freely within a budget, which would be sound and far too large, but the outcome of an inner problem whose own reachable set is what the jump may be. The next lemma says that this problem is well posed and bounds it in closed form.

Lemma 3.2 (The bubble is a single arch). Assume [Assumptions 2.1](#) and [2.2](#) and let $v^- \leq -v_0 < 0$ with $|v^-| \leq v_{\max}$. Along every inner trajectory from (v^-, s^-) :

(a) v is strictly increasing and ζ is strictly concave, with $\zeta > 0$ on $(0, I)$ and

$$\frac{2|v^-|}{L_{\max}} \leq I \leq \frac{2|v^-|}{L_{\min}} \leq I_{\max} := \frac{2v_{\max}}{L_{\min}};$$

(b) ζ attains its maximum where v vanishes, and

$$\frac{(v^-)^2}{2L_{\max}} \leq \max_{[0, I]} \zeta \leq \frac{(v^-)^2}{2L_{\min}};$$

(c) $\sqrt{L_{\min}/L_{\max}} |v^-| \leq v(I) \leq \sqrt{L_{\max}/L_{\min}} |v^-|$;

(d) $|s(I) - s^-| \leq \Gamma_{\max} I \leq \Gamma_{\max} I_{\max}$.

Proof. (a) By [Assumption 2.2](#), $dv/d\mu = L \geq L_{\min} > 0$, so v increases strictly, and $d^2\zeta/d\mu^2 = -L \leq -L_{\min}$, so ζ is strictly concave. At $\mu = 0$ we have $\zeta = 0$ and $d\zeta/d\mu = -v^- = |v^-| > 0$, so ζ increases initially. Integrating $d^2\zeta/d\mu^2 \leq -L_{\min}$ twice from these data gives $\zeta(\mu) \leq |v^-|\mu - \frac{1}{2}L_{\min}\mu^2$, which vanishes at $2|v^-|/L_{\min}$, whence the upper bound on I ; the lower bound follows in the same way from $\zeta(\mu) \geq |v^-|\mu - \frac{1}{2}L_{\max}\mu^2$.

(b) The maximum is where $d\zeta/d\mu = -v$ vanishes. On $[0, I]$ regard $\frac{1}{2}v^2$ as a function of ζ along either branch: since $d\zeta/d\mu = -v$ and $dv/d\mu = L$,

$$\frac{d}{d\zeta} \left(\frac{1}{2}v^2 \right) = v \frac{dv}{d\zeta} = v \cdot \frac{L}{-v} = -L. \quad (3.5)$$

Integrating [Eq. \(3.5\)](#) along the ascending branch, from $\zeta = 0$ where $v = v^-$ to $\zeta = \max \zeta$ where $v = 0$, gives $\frac{1}{2}(v^-)^2 = \int_0^{\max \zeta} L d\zeta$, and L lies between $L_{\min}$ and $L_{\max}$.

(c) Integrating Eq. (3.5) along the descending branch, from $\zeta = \max \zeta$ where $v = 0$ to $\zeta = 0$ where $v = v(I)$, gives $\frac{1}{2}v(I)^2 = \int_0^{\max \zeta} L d\zeta$ as well. Both $\frac{1}{2}(v^-)^2$ and $\frac{1}{2}v(I)^2$ therefore lie in $[L_{\min} \max \zeta, L_{\max} \max \zeta]$, and the claim is the ratio of the two bounds. In particular I is finite and the exit data are attained.

(d) $s(I) - s^- = \int_0^I \Gamma d\mu$, and $|\Gamma| \leq \Gamma_{\max}$. $\square$

Part (c) is worth a word. For a layer force of constant strength the exit speed equals the entry speed: the layer turns the trajectory without doing net work on the altitude channel, exactly as lift does no work. The spread $\sqrt{L_{\max}/L_{\min}}$ is the room the control has, and it is what makes J a set and not a point.

Lemma 3.3 (The pass map is Lipschitz). *Assume Assumptions 2.1 and 2.2 and fix $v_0 > 0$. Put $\kappa_1 = \kappa_L + \kappa_\Gamma$ and $v_e = \sqrt{L_{\min}/L_{\max}} v_0$. Then for all $v_1^-, v_2^- \leq -v_0$ with $|v_i^-| \leq v_{\max}$ and all $s_1^-, s_2^- \in S$,*

$$d_H\left(J(v_1^-, s_1^-), J(v_2^-, s_2^-)\right) \leq \text{Lip}_J (|v_1^- - v_2^-| + |s_1^- - s_2^-|),$$

where d_H is the Hausdorff distance and

$$\text{Lip}_J = e^{\kappa_1 I_{\max}} \left(1 + \frac{(L_{\max} + \Gamma_{\max}) I_{\max}}{v_e}\right). \quad (3.6)$$

Proof. Fix (c, θ) and compare the two inner trajectories it generates. In Eq. (3.4) the equations for v and s do not involve ζ , and their right-hand sides depend only on s , with Lipschitz constant κ_1 ; so by Gronwall's inequality, writing $\Delta = |\Delta v| + |\Delta s|$,

$$\Delta(\mu) \leq e^{\kappa_1 \mu} \Delta(0) \quad \text{for } 0 \leq \mu \leq \min(I_1, I_2) \leq I_{\max}.$$

Since $\Delta\zeta(\mu) = -\int_0^\mu \Delta v$, this gives $|\Delta\zeta(\mu)| \leq I_{\max} e^{\kappa_1 I_{\max}} \Delta(0)$. For the exit times, suppose $I_1 \leq I_2$. By Lemma 3.2(a) and (c) the function ζ_2 is concave with $\zeta_2(I_2) = 0$ and $d\zeta_2/d\mu = -v_2(I_2) \leq -v_e$ there, so $\zeta_2(\mu) \geq v_e(I_2 - \mu)$ for $\mu \leq I_2$. Taking $\mu = I_1$, where $\zeta_1 = 0$, gives $v_e(I_2 - I_1) \leq \zeta_2(I_1) = \Delta\zeta(I_1)$, that is $|I_2 - I_1| \leq |\Delta\zeta(I_1)|/v_e$. Finally $|dv/d\mu| + |ds/d\mu| \leq L_{\max} + \Gamma_{\max}$, so carrying trajectory 2 from I_1 to I_2 moves its exit data by at most $(L_{\max} + \Gamma_{\max})|I_2 - I_1|$. Adding the two contributions gives the bound with the constant Eq. (3.6), for each fixed (c, θ) ; taking suprema over (c, θ) on each side turns it into the stated bound on the Hausdorff distance. $\square$

4 The limit system, and the theorem

The skeleton and the bubbles together are not a surface with corrections but a system of its own, with a flow and a jump, and we state the target against it.

Definition 4.1 (The limit system). Fix $v_0 > 0$. The limit system $\mathcal{H}$ has state (y, v, s) and two transitions.

Arc (flow). $\dot{y} = v$, $\dot{v} = a(s, c, \theta)$, $\dot{s} = F(s, c, \theta)$, while $y \geq 0$.

Bubble (jump). At $y = 0$ with $v \leq -v_0$, the state $(0, v^-, s^-)$ is replaced by $(0, v^+, s^+)$ with $(v^+, s^+) \in J(v^-, s^-)$, consuming no time.

For $t \geq 0$ let $\mathcal{R}_{\mathcal{H}}(t)$ be the set of states reached at time t from $\mathcal{X}_0$ over all measurable $c(\cdot), \theta(\cdot)$ and all selections from J , and write $\mathcal{R}_{\mathcal{H}}(E) = \bigcup_{t \in E} \mathcal{R}_{\mathcal{H}}(t)$.

The arc is the skeleton: the motion where the layer is not felt. The jump is the bubble, and it is a jump of the velocity and of the slow state at a fixed altitude. The limit system carries no ε . We measure the residual in the norm $|(y, v, s)| = |y| + |v| + |s|$ on $\mathbb{R} \times \mathbb{R} \times \mathbb{R}^d$ and write $\mathbb{B}(\varrho)$ for the ball of radius ϱ in it, and $\oplus$ for the Minkowski sum. Finally $\mathcal{R}_\varepsilon(T)$ denotes the set of states reached at time T by Eqs. (2.2) to (2.4) from $\mathcal{X}_0$, over every admissible control and every selection.

Theorem 4.2 (Outer inclusion with an explicit residual). *Assume Assumptions 2.1 to 2.4. Let*

$$v_{\min} = \frac{v_*}{\sqrt{2}}, \quad v_e = \sqrt{\frac{L_{\min}}{L_{\max}}} v_{\min}, \quad N_* = \left\lfloor \frac{ML_{\max}}{2v_*} \right\rfloor + 1, \quad h_* \text{ as in Eq. (5.9),} \quad (4.1)$$

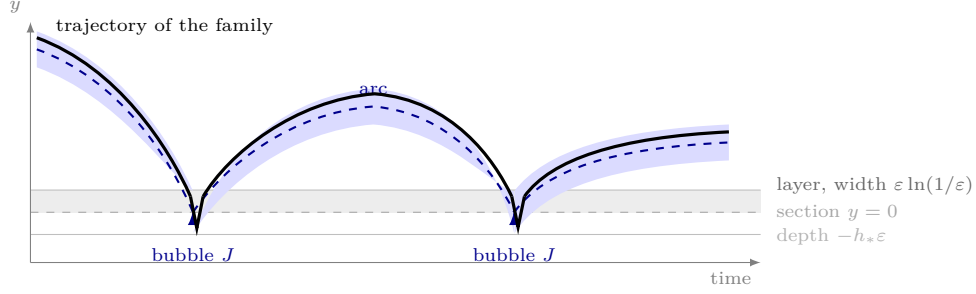


Figure 4.1: [Theorem 4.2](#), as a schematic. A trajectory of [Eqs. \(2.2\) to \(2.4\)](#) (solid) descends, rebounds inside the layer in a time of order ε , and climbs again; the trajectory of the limit system (dashed) flows along the arcs and jumps at the section, by the pass map J of [Definition 3.1](#). The two stay within the residual (shaded). The layer and the depth are drawn far thicker than they are: both shrink with ε while the arcs do not, which is the whole of the difficulty. Nothing here is computed.

and build $\mathcal{H}$ with $v_0 = v_{\min}$. There are $\varepsilon_1 > 0$ and $C < \infty$, both given explicitly in terms of the constants of [Assumptions 2.1 to 2.4](#) by [Eqs. \(5.15\) and \(5.16\)](#) below, such that for every $\varepsilon \in (0, \varepsilon_1]$, writing $\varrho(\varepsilon) = C\varepsilon \ln(1/\varepsilon)$,

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\mathcal{R}_{\mathcal{H}}([T - \varrho(\varepsilon), T + \varrho(\varepsilon)])}_{\text{skeleton and bubbles}} \oplus \underbrace{\mathbb{B}(\varrho(\varepsilon))}_{\text{residual}}. \quad (4.2)$$

Moreover every trajectory of [Eqs. \(2.2\) to \(2.4\)](#) makes at most N_* passes, spends a time at most $N_*(2\ln(1/\varepsilon)/v_{\min} + M)\varepsilon$ with $y \leq \varepsilon \ln(1/\varepsilon)$, and satisfies $y \geq -h_*\varepsilon$ throughout.

Three things about the statement. The inclusion is over a short interval of times rather than at the instant T , of half-width the residual itself: that is the price of comparing a pass of positive duration with an instantaneous jump, and the two clocks may slip either way, since the limit system spends no time in the pass but descends further to reach the section. The constant C depends on none of the control, the selection or the initial point, by [Remark 2.5](#), so the bound is a statement about the whole admissible family. And the residual is $O(\varepsilon \ln(1/\varepsilon))$ rather than $O(\varepsilon)$: the logarithm is the price of localising an exponential layer at a single section, and we do not claim it is sharp. Removing it is a small, concrete first problem, and it asks for a first-moment correction to the matching, since the impulse-weighted mean altitude of the layer is of order ε and not of order $\varepsilon \ln(1/\varepsilon)$.

5 Proof

Fix $\varepsilon \in (0, \varepsilon_1]$, where ε_1 is specified in [Eq. \(5.16\)](#) below, and fix a trajectory $x_\varepsilon = (y, v, s)$ of [Eqs. \(2.2\) to \(2.4\)](#) on $[0, T]$ with control $c(\cdot)$, selection $\theta(\cdot)$ and $x_\varepsilon(0) \in \mathcal{X}_0$. Throughout, put

$$\Lambda = \ln(1/\varepsilon), \quad \text{so that} \quad e^{-\Lambda} = \varepsilon \quad \text{and} \quad \varepsilon e^\Lambda = 1. \quad (5.1)$$

The level $y = \Lambda\varepsilon$ is the edge of the layer: above it the density is at most $\varepsilon^{-1}e^{-\Lambda} = 1$, and the proof consists in showing that the impulse delivered above that level is small while the time spent below it is short.

Step 1: the budget

By [Assumption 2.3](#), along x_ε ,

$$\frac{d}{dt}\Theta_d(s) = \nabla\Theta_d \cdot F + \rho_\varepsilon(y)\nabla\Theta_d \cdot \Gamma \leq -c_0\rho_\varepsilon(y).$$

Integrating over $[0, T]$ and using [Eq. \(2.5\)](#),

$$\int_0^T \rho_\varepsilon(y(t)) dt \leq \frac{\Theta_d(s(0)) - \Theta_d(s(T))}{c_0} \leq M. \quad (5.2)$$

This holds for every control, every selection and every ε .

Step 2: how many passes

Let $P = (t_a, t_b)$ be a pass with $t_b < T$. Then $y(t_a) = y(t_b) = \ell_1$, and since the trajectory enters $\{y < \ell_1\}$ at t_a and leaves at t_b we have $v(t_a) \leq 0 \leq v(t_b)$; with [Assumption 2.4\(ii\)](#) this gives $v(t_a) \leq -v_*$ and $v(t_b) \geq v_*$. Integrating [Eq. \(2.3\)](#) over P and using $a \leq 0$ from [Assumption 2.2](#),

$$2v_* \leq v(t_b) - v(t_a) = \int_P (a + \rho_\varepsilon(y)L) dt \leq L_{\max} \int_P \rho_\varepsilon(y) dt,$$

so each complete pass consumes at least $2v_*/L_{\max}$ of the budget [Eq. \(5.2\)](#). Counting a possible incomplete final pass, the number N of passes satisfies

$$N \leq \left\lfloor \frac{ML_{\max}}{2v_*} \right\rfloor + 1 = N_*. \quad (5.3)$$

The passes are therefore finite in number, with a bound independent of ε , and this is the only place where finiteness is used.

Step 3: speed at the edge of the layer

Suppose $\varepsilon \leq v_*^2/(4L_{\max})$, so that by [Eq. \(5.1\)](#) $L_{\max}e^{-\Lambda} \leq v_*^2/4$. We claim that on each pass,

$$|v| \geq v_{\min} = \frac{v_*}{\sqrt{2}} \quad \text{whenever} \quad \Lambda\varepsilon \leq y \leq \ell_1. \quad (5.4)$$

Let $P = (t_a, t_b)$ be a pass. On any subinterval where v does not vanish, y is strictly monotone and may be used as the independent variable; since $dy = v dt$, dividing [Eq. \(2.3\)](#) by v gives there

$$\frac{1}{2} \frac{d(v^2)}{dy} = a + \rho_\varepsilon(y)L \leq \rho_\varepsilon(y)L_{\max}, \quad (5.5)$$

the inequality by [Assumption 2.2](#), and this holds whatever the sign of v . Integrating [Eq. \(5.5\)](#) upward in y from any level y' to ℓ_1 , and using $\int_{y'}^{\ell_1} \rho_\varepsilon(y) dy = e^{-y'/\varepsilon} - e^{-\ell_1/\varepsilon} \leq e^{-y'/\varepsilon}$, gives the single estimate

$$\frac{1}{2}v(\ell_1)^2 - \frac{1}{2}v(y')^2 \leq L_{\max}e^{-y'/\varepsilon}, \quad (5.6)$$

valid along any branch of the pass on which v keeps one sign between the level y' and the section $y = \ell_1$.

Now suppose, for contradiction, that v vanishes at some $t_c \in P$ with $y_c := y(t_c) \geq \Lambda\varepsilon$, and take t_c minimal. On $[t_a, t_c]$ we have $v < 0$, so [Eq. \(5.6\)](#) applies with $y' = y_c$ and $v(\ell_1) = v(t_a)$. Since $v(t_c) = 0$ and $y_c \geq \Lambda\varepsilon$ it reads

$$\frac{1}{2}v_*^2 \leq \frac{1}{2}v(t_a)^2 \leq L_{\max}e^{-\Lambda} \leq \frac{1}{4}v_*^2,$$

a contradiction. Hence v does not vanish while $y \geq \Lambda\varepsilon$, and [Eq. \(5.6\)](#) applied at any level $y' = y \in [\Lambda\varepsilon, \ell_1]$ of that descent gives

$$\frac{1}{2}v(y)^2 \geq \frac{1}{2}v(t_a)^2 - L_{\max}e^{-\Lambda} \geq \frac{1}{2}v_*^2 - \frac{1}{4}v_*^2 = \frac{1}{4}v_*^2,$$

which is [Eq. \(5.4\)](#). By [Assumption 2.4\(iii\)](#) the function v vanishes at most once on P , so $v < 0$ before that time and $v > 0$ after it; on the ascending branch [Eq. \(5.6\)](#) applies in the same way, with $v(\ell_1) = v(t_b)$ and $|v(t_b)| \geq v_*$, and yields [Eq. \(5.4\)](#) again. In particular y decreases and then increases on P : a pass is one dip, and

$$W = \{t \in P : y(t) \leq \Lambda\varepsilon\} \quad (5.7)$$

is a single interval $[\alpha, \beta]$ with $y(\alpha) = y(\beta) = \Lambda\varepsilon$. We call it the *window* of the pass.

Step 4: the window is short and shallow, and little is delivered outside it

(a) *Duration.* Split W at the section. On $W \cap \{0 \leq y \leq \Lambda\varepsilon\}$ the altitude is monotone on each of the two branches and $|v| \geq v_{\min}$ by Eq. (5.4), so the time spent there is at most $2\Lambda\varepsilon/v_{\min}$. On $W \cap \{y < 0\}$ we have $\rho_\varepsilon(y) \geq \varepsilon^{-1}$, so by Eq. (5.2) the time spent there is at most $M\varepsilon$. Hence

$$|W| \leq \left(\frac{2\Lambda}{v_{\min}} + M \right) \varepsilon. \quad (5.8)$$

(b) *Depth.* Let $y_{\min}$ be the altitude where v vanishes on P . Returning to the identity in Eq. (5.5) and bounding it below by $-a_{\max} + \rho_\varepsilon(y)L_{\min}$ instead, then integrating from $y_{\min}$ to ℓ_1 ,

$$\frac{1}{2}v(t_a)^2 \geq -a_{\max}(\ell_1 - y_{\min}) + L_{\min}(e^{-y_{\min}/\varepsilon} - e^{-\ell_1/\varepsilon}),$$

so that, using $|v| \leq v_{\max}$, $|y_{\min}| \leq y_{\max}$ and $e^{-\ell_1/\varepsilon} \leq 1$,

$$y_{\min} \geq -h_*\varepsilon, \quad h_* = \ln\left(1 + \frac{\frac{1}{2}v_{\max}^2 + a_{\max}(\ell_1 + y_{\max})}{L_{\min}}\right). \quad (5.9)$$

In the other direction, Eq. (5.6) at the level $y' = y_{\min}$, where v vanishes, gives $\frac{1}{2}v_*^2 \leq L_{\max}e^{-y_{\min}/\varepsilon}$, that is $y_{\min} \leq \varepsilon \ln(2L_{\max}/v_*^2)$, so the turning point lies inside the window as soon as $\Lambda > \ln(2L_{\max}/v_*^2)$.

(c) *Impulse outside the windows.* Let $O = [0, T] \setminus \bigcup_k W_k$. On $O \cap \{y \geq \ell_1\}$ we have $\rho_\varepsilon(y) \leq \varepsilon^{-1}e^{-\ell_1/\varepsilon}$. The set $O \cap \{\Lambda\varepsilon \leq y < \ell_1\}$ consists of at most $2N$ branches on which y is monotone, by Step 3, and on each of them $|v| \geq v_{\min}$, so $dt = dy/|v|$ and

$$\int_{O \cap \{\Lambda\varepsilon \leq y < \ell_1\}} \rho_\varepsilon(y) dt \leq \frac{2N}{v_{\min}} \int_{\Lambda\varepsilon}^{\ell_1} \rho_\varepsilon(y) dy \leq \frac{2N_*}{v_{\min}} e^{-\Lambda}.$$

Hence, with Eq. (5.1),

$$\sigma(\varepsilon) := \int_O \rho_\varepsilon(y(t)) dt \leq \frac{2N_*}{v_{\min}} \varepsilon + T\varepsilon^{-1}e^{-\ell_1/\varepsilon} \leq \left(\frac{2N_*}{v_{\min}} + 1 \right) \varepsilon, \quad (5.10)$$

the last step holding once ε is small enough that $T\varepsilon^{-1}e^{-\ell_1/\varepsilon} \leq \varepsilon$. This is the step that makes the decomposition possible: although the total impulse along the trajectory is of order one, all but $O(\varepsilon)$ of it is delivered inside the windows, whose total duration is $O(\varepsilon \ln(1/\varepsilon))$ by Eq. (5.8).

Step 5: a window is a bubble

Fix a window $W = [\alpha, \beta]$ and let μ and ζ be as in Eq. (3.1), with $t_0 = \alpha$. Then $\zeta(\alpha) = \zeta(\beta) = e^{-\Lambda} = \varepsilon$, the μ -length of the window is $I_\varepsilon = \int_W \rho_\varepsilon dt \leq M$ by Eq. (5.2), and Eq. (3.2) holds exactly. Let $(\tilde{v}, \tilde{\zeta}, \tilde{s})$ be the inner trajectory of Eq. (3.4) driven by the *same* (c, θ) , read as functions of μ and extended arbitrarily beyond I_ε , started at $(\tilde{\zeta}, \tilde{v}, \tilde{s})(0) = (0, v(\alpha), s(\alpha))$, with exit time I . Note that $v(\alpha) \leq -v_{\min}$ by Eq. (5.4), so Lemma 3.2 applies with $v_0 = v_{\min}$. Write $M_* = \max(M, I_{\max})$.

Subtracting Eq. (3.4) from Eq. (3.2), the difference of the (v, s) components satisfies

$$\left| \frac{d}{d\mu} (\Delta v, \Delta s) \right| \leq \kappa_1 |\Delta s| + \frac{\varepsilon}{\zeta} (a_{\max} + F_{\max}),$$

and by Eq. (3.3) the integral of the last term over the window is exactly $(a_{\max} + F_{\max})|W|$. Since the difference vanishes at $\mu = 0$, Gronwall's inequality and Eq. (5.8) give, on $[0, \min(I_\varepsilon, I)]$,

$$|\Delta v| + |\Delta s| \leq e_1(\varepsilon) := e^{\kappa_1 M_*} (a_{\max} + F_{\max}) \left(\frac{2\Lambda}{v_{\min}} + M \right) \varepsilon. \quad (5.11)$$

For the rescaled density, $\Delta\zeta(\mu) = \varepsilon - \int_0^\mu \Delta v$, so $|\Delta\zeta| \leq \varepsilon + M_* e_1(\varepsilon)$. The two exit conditions differ: $\tilde{\zeta}(I) = 0$ while $\zeta(\beta) = \varepsilon$. Arguing exactly as in Lemma 3.3, with $\tilde{\zeta}$ concave and $|d\tilde{\zeta}/d\mu| \geq v_e$ at its exit,

$$|I_\varepsilon - I| \leq \frac{|\Delta\zeta(I_\varepsilon)| + \varepsilon}{v_e} \leq \frac{2\varepsilon + M_* e_1(\varepsilon)}{v_e}.$$

Finally, by Eqs. (3.2) and (5.1) and $\zeta \geq \varepsilon$ on the window, $|dv/d\mu| + |ds/d\mu| \leq L_{\max} + \Gamma_{\max} + a_{\max} + F_{\max} =: \Pi_*$, so carrying either trajectory between I_ε and I moves its data by at most $\Pi_*|I_\varepsilon - I|$. Combining,

$$(v(\beta), s(\beta)) \in J(v(\alpha), s(\alpha)) \oplus \mathbb{B}(e_2(\varepsilon)), \quad e_2(\varepsilon) = e_1(\varepsilon) + \frac{\Pi_*(2\varepsilon + M_*e_1(\varepsilon))}{v_e}, \quad (5.12)$$

while $y(\alpha) = y(\beta) = \Lambda\varepsilon$. A window is therefore a bubble, up to $e_2(\varepsilon)$ in the velocity and the slow state and up to $\Lambda\varepsilon$ in the altitude, and both are $O(\varepsilon \ln(1/\varepsilon))$.

Step 6: assembling the comparison

Let $W_1, \dots, W_N$ be the windows, $W_k = [\alpha_k, \beta_k]$, and let $O_0, \dots, O_N$ be the complementary intervals. Construct a trajectory $x_{\mathcal{H}}$ of $\mathcal{H}$ as follows: start at $x_\varepsilon(0)$; follow the arc flow with the same (c, θ) ; at each hit of the section $y = 0$, jump, choosing the point of J at the entry data of $x_{\mathcal{H}}$ nearest to $(v(\beta_k), s(\beta_k))$; continue. Write Δ_k for the distance between $x_\varepsilon(\beta_k)$ and $x_{\mathcal{H}}$ just after its k -th jump, and $\Delta_0 = 0$.

On an arc, x_ε and $x_{\mathcal{H}}$ obey the same field f_0 , which is Lipschitz on $\mathcal{X}$ with constant $\kappa_0 = 1 + \kappa_a + \kappa_F$, except that x_ε carries in addition $\rho_\varepsilon(y)b$ with $|b| \leq b_{\max} := L_{\max} + \Gamma_{\max}$. Gronwall's inequality over O_k and Eq. (5.10) therefore give, at the end of the arc, a distance at most $e^{\kappa_0 T}(\Delta_k + b_{\max} \int_{O_k} \rho_\varepsilon dt)$. At the next window, Eq. (5.12) places $(v(\beta_{k+1}), s(\beta_{k+1}))$ within $e_2(\varepsilon)$ of J evaluated at x_ε 's own entry data, and Lemma 3.3 transfers this to J evaluated at $x_{\mathcal{H}}$'s entry data at the cost of the factor Lip_J ; the altitudes differ by at most $\Lambda\varepsilon$. Hence, with $A = e^{\kappa_0 T} \max(1, \text{Lip}_J)$,

$$\Delta_{k+1} \leq A \left(\Delta_k + b_{\max} \int_{O_k} \rho_\varepsilon dt \right) + e_2(\varepsilon) + \Lambda\varepsilon.$$

Iterating over at most N_* windows and one final arc, and using $\sum_k \int_{O_k} \rho_\varepsilon dt = \sigma(\varepsilon)$ from Eq. (5.10),

$$|x_\varepsilon(T) - x_{\mathcal{H}}(\hat{T})| \leq A^{N_*+1} \left[b_{\max} \sigma(\varepsilon) + N_*(e_2(\varepsilon) + \Lambda\varepsilon) \right], \quad (5.13)$$

where $\hat{T}$ is the time at which $x_{\mathcal{H}}$ arrives, which differs from T because x_ε spends the windows where $x_{\mathcal{H}}$ spends no time at all and reaches the section from a slightly different altitude. By Eqs. (5.4) and (5.8) that discrepancy is at most

$$|T - \hat{T}| \leq N_* \left(\frac{4\Lambda}{v_{\min}} + M \right) \varepsilon. \quad (5.14)$$

It remains to collect constants. Since $\Lambda \geq 1$, Eqs. (5.11) and (5.12) give $e_2(\varepsilon) + \Lambda\varepsilon \leq c_2 \Lambda\varepsilon$ with

$$c_2 = \left(1 + \frac{M_* \Pi_*}{v_e} \right) e^{\kappa_1 M_*} (a_{\max} + F_{\max}) \left(\frac{2}{v_{\min}} + M \right) + \frac{2\Pi_*}{v_e} + 1,$$

and Eq. (5.10) gives $\sigma(\varepsilon) \leq (2N_*/v_{\min} + 1)\Lambda\varepsilon$. Substituting into Eqs. (5.13) and (5.14) and recalling $\Lambda = \ln(1/\varepsilon)$, both the state discrepancy and the time discrepancy are at most $C\varepsilon \ln(1/\varepsilon)$ with

$$C = A^{N_*+1} \left[b_{\max} \left(\frac{2N_*}{v_{\min}} + 1 \right) + N_* c_2 \right] + N_* \left(\frac{4}{v_{\min}} + M \right), \quad (5.15)$$

which is the constant promised in Theorem 4.2. The admissible range of ε is

$$\varepsilon_1 = \min \left\{ e^{-1}, \frac{v_*^2}{4L_{\max}}, \varepsilon', \varepsilon'' \right\}, \quad (5.16)$$

where ε' is the largest $\varepsilon \leq e^{-1}$ with $\varepsilon \ln(1/\varepsilon) \leq \min\{\ell_1/2, y_{\max}/h_*\}$ and $\ln(1/\varepsilon) > \ln(2L_{\max}/v_*^2)$, and ε'' the largest with $T\varepsilon^{-1}e^{-\ell_1/\varepsilon} \leq \varepsilon$; these are the conditions used in Steps 3, 4(b) and 4(c). Since $x_\varepsilon(T)$ was an arbitrary point of $\mathcal{R}_\varepsilon(T)$, since $x_{\mathcal{H}}(\hat{T}) \in \mathcal{R}_{\mathcal{H}}(\hat{T})$, and since Eq. (5.14) places $\hat{T}$ in $[T - \varrho(\varepsilon), T + \varrho(\varepsilon)]$, Eq. (5.13) is exactly Eq. (4.2). The claims about the pass count, the time spent in the layer and the depth are Eqs. (5.3), (5.8) and (5.9). $\square$

Remark 5.1 (Where the hypotheses were used). [Assumption 2.3](#) was used once, in Step 1, and everything quantitative descends from it: the pass count, the time below the section, and the μ -horizon of Step 5. [Assumption 2.2](#) gave the sign of a in Step 2, the concavity of ζ in [Lemma 3.2](#), and the lower bound on the exit speed that keeps J Lipschitz. [Assumption 2.4](#) was used only in Step 3, to make the window a single interval and to supply the margin v_* that [Eq. \(5.4\)](#) turns into a speed bound at the edge of the layer. Nothing else is needed; in particular no compactness argument and no measure appears anywhere above.

6 Where the occupation measures enter

[Theorem 4.2](#) is a statement about trajectories and its proof uses no measures. That is worth saying plainly, because the program was conceived the other way round. The measures are not what crosses the singular limit; they are what certifies the reduced problem on the far side of it. We give the definitions, so that what is open about them can be stated precisely.

For a trajectory of [Eqs. \(2.2\) to \(2.4\)](#) with control $c(\cdot)$ and selection $\theta(\cdot)$, the *occupation measure* is the element $\mu_\varepsilon \in \mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta)$ defined by

$$\int \varphi \, d\mu_\varepsilon = \int_0^T \varphi(t, x_\varepsilon(t), c(t), \theta(t)) \, dt \quad \text{for } \varphi \in C([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta),$$

together with the initial and terminal measures μ_0, μ_T carried by $x_\varepsilon(0)$ and $x_\varepsilon(T)$. The topology is the weak-* topology, that is, the topology of the dual of that space of continuous functions; the base is a compact metric space by [Assumption 2.1](#), so the topology is metrizable and bounded sets are relatively compact. In these terms the dynamics are a *linear* constraint: for every $\varphi \in C^1$,

$$\int \varphi(T, \cdot) \, d\mu_T - \int \varphi(0, \cdot) \, d\mu_0 = \int \left(\frac{\partial \varphi}{\partial t} + \nabla \varphi \cdot (f_0 + \rho_\varepsilon b) \right) \, d\mu_\varepsilon, \quad (6.1)$$

which is the Liouville equation of the family, and the reachable set at time T is the union of the supports of the terminal measures that satisfy it. This is the occupation-measure program of Lasserre, Henrion and collaborators [1, 2, 3], and an outer bound comes from its dual: a pair (Φ, Ψ) with Φ non-decreasing along the admissible motions and at least one on $\mathcal{X}_0$, and Ψ majorising Φ at time T , encloses the reachable set in $\{\Psi \geq 1\}$.

What [Theorem 4.2](#) does is to hand this machinery a different problem. After the theorem, the object to be enclosed is $\mathcal{R}_\mathcal{H}$, which is a reachability problem in $2 + d$ coordinates with a flow, one guard $y = 0$, a set-valued reset J , and at most N_* jumps. That is exactly the hybrid form for which the occupation-measure relaxation is known [4, 5]: a measure on each guard carries the mass from one mode to the next, and the certificate Φ must be non-decreasing along the arc flow and across the reset. The singular limit has been discharged by [Section 5](#), which is elementary; the dimension and the provenance of the guarantee are discharged on $\mathcal{H}$, which is low-dimensional by construction. The division of labour is the point.

Two things are not done. [Theorem 4.2](#) compares trajectories and not measures, and the companion statement at the level of measures is [objective 4](#). And nothing above makes $\mathcal{R}_\mathcal{H}$ tight: an outer bound on it is an outer bound on $\mathcal{R}_\varepsilon(T)$ up to the residual, by [Eq. \(4.2\)](#), but how much is lost in passing from the one to the other is a separate question from how much is lost in the limit.

7 What hypersonic entry adds

The application is hypersonic entry: the reachable set of an entry vehicle, for any vehicle and any corridor, with the vehicle and its corridor entering only as data. Entry is not an arbitrary illustration, because it supplies the singular perturbation, the limit system and the concentrating events in one system. Nor is it already covered: the exact method solves a Hamilton–Jacobi–Isaacs equation on a grid, capped in practice near four or five states [6, 7], while a rigid-body entry vehicle carries thirteen and more once ablation and the aerothermoelastic modes are coupled [8]; and set-propagation methods return rigorous enclosures at ten to twenty states [9, 10, 11], but of one instance — one vehicle, one initial box, one horizon — with no explicit dependence on the corridor or on the width of the aerodynamic uncertainty, which is what a statement uniform over a family requires.

7.1 The correspondence

Measure altitude on the planetary length scale, $y = h/R_e$, so that an exponential atmosphere of scale height H_s has density proportional to $e^{-y/\varepsilon}$ with

$$\varepsilon = H_s/R_e,$$

the thin-atmosphere ratio. Then ρ_ε of Eq. (2.1) is the atmospheric density, normalised as in Eq. (1.1); v is the rate of climb; s collects the speed, the range, the heading and the energy; a is minus the net gravitational acceleration $g - V^2/r$ that the lift must balance; L is the vertical component of the lift per unit mass divided by the density; and Γ carries the drag. The selection θ is a choice of aerodynamic coefficients from an enclosure, so that Eqs. (2.2) to (2.4) are a differential inclusion rather than a single field [12, 13], and Remark 2.5 is the statement that the bound holds over the whole enclosure.

Assumption 2.3 is the hypothesis that transfers exactly. The specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, satisfies $\dot{\mathcal{E}} = -P$ with $P = DV/m \geq 0$ the drag power, because neither the lift nor the Coriolis force does work. Taking $\Theta_d = \mathcal{E}$ gives $\nabla \Theta_d \cdot F = 0$ exactly, and $\nabla \Theta_d \cdot \Gamma \leq -c_0$ with c_0 fixed by lower bounds on the drag coefficient and the speed, since the drag is strictly proportional to the density. The budget, the pass count and the bound on the time spent in the layer therefore hold for entry as they stand. Assumption 2.2 holds on a corridor bounded away from circular speed, where $g - V^2/r$ is bounded below, and with the bank bounded away from ninety degrees; Assumption 2.1 holds away from the geographic poles, where the spherical chart is singular.

7.2 Why Assumption 2.4 fails

Assumption 2.4 is the hypothesis entry does not satisfy, and the reason is quantitative. A pass entered at a flight-path angle γ_e crosses a scale height in a time of order ε/γ_e on the slow time scale and levels out under a load of order γ_e^2/ε relative to gravity, so the entry angle enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman–Loh variables [14, 15]. For the Earth $\sqrt{\varepsilon} \approx 0.034$, about two degrees, and entry angles are a few degrees. A pass is therefore neither short compared with the arc that follows it nor distinct in scale from the oscillation about the equilibrium glide, and the margin v_* of Assumption 2.4(ii) is not available uniformly.

8 Plan of work

Theorem 4.2 is the first of the obligations, and it is discharged. Four remain. We state each with the physical obstruction it has to overcome.

Objective 1. Grazing and capture: the glide as a second flow

Assumption 2.4 excludes a trajectory whose passes weaken and accumulate, and that is precisely capture into the equilibrium glide, on which lift balances weight less centrifugal relief and the vehicle flies for the rest of the entry; in the reduced model it is the limit $v_* \rightarrow 0$, where the counting argument of Step 2 gives nothing. We must remove Assumption 2.4, so that the skeleton carries both the arcs and the glide and the transition between them lies inside the limit system rather than in the residual. The budget Eq. (5.2) survives unchanged; what is lost is only the pass count Eq. (5.3). Passes that dissipate at least a threshold δ are still at most M/δ in number, so the obligation is to show that what falls below the threshold is an oscillation about the glide, to be averaged rather than resolved, with δ chosen against the residual it leaves behind. The glide is not a normally hyperbolic slow manifold — the oscillation normal to it has a frequency growing like $\varepsilon^{-1/2}$ and a damping rate independent of ε — so no boundary-layer estimate of Tikhonov type brings a trajectory onto it [16], which agrees with the existing regular-perturbation analysis of the glide [17]. The objective is met by a theorem of the form of Theorem 4.2 with Assumption 2.4 replaced by such a threshold, the residual still explicit.

Objective 2. The bubble about a moving section

In Eqs. (2.2) to (2.4) the section $y = 0$ is fixed. In entry the level about which the vehicle oscillates is the glide altitude, and a change of bank moves it by a fixed fraction of a scale height however thin the layer; the amplitude is then a coordinate on which the control acts, since a bank history in step with the oscillation pumps it just as one opposed to it damps it. Guidance laws use the latter [18] and a reachable set must allow for the former, so the amplitude belongs to the limit system and not to the residual. We must therefore pose the inner problem of Definition 3.1 about an equilibrium that the control moves, and re-prove Lemmas 3.2 and 3.3 there. What has to be re-examined is the change of variable Eq. (3.1), since it is the exactness of $d\zeta/d\mu = -v$ that makes the inner system free of ε .

Objective 3. Separation of scales under a bounded selection family

The reduced model has one fast direction; an entry vehicle has others. The attitude and the elastic modes oscillate for as long as there is air and nothing about them concentrates, so in a profile decomposition they are not profiles at all but belong to the remainder, which is not small in the conserved quantity [19]. The tool for them is averaging [20, 21], and two things stand between that tool and a bound: an averaging theorem gives the error to an order, and an order is not a bound; and with several oscillations it holds only away from resonance, while the attitude frequency grows with the square root of the dynamic pressure and so rises from zero at the start of every pass. The results that carry a solution through a resonance without resolving it hold, where capture is possible, outside a set of initial conditions of small measure — and an outer bound can discard nothing. We must prove an averaging theorem with constants, uniform over the selections and valid for every trajectory; where two bands cross, the statement must be for the coupled pair, carried jointly through the crossing. **This is the critical path.** It is the one obligation for which no theory is at hand, and the part of the program for which I am looking for an advisor.

Objective 4. The measure-level companion and the certificate

We must show that the occupation measures of Section 6 converge to measures satisfying the Liouville equation of the limit system, with the mass crossing each pass carried by a measure on the graph of J , uniformly over the admissible family, and exhibit the dual pair (Φ, Ψ) non-decreasing along the arc flow and across the reset. This is what turns Theorem 4.2 into a certificate rather than an approximation theorem, and it is where the machinery the program was conceived around — weak-* compactness, and concentration-compactness in the form that resolves each profile at its own scale [22, 23] — is expected to be needed: for the structure, not for the constants.

8.1 Sequencing

Three phases rather than four parallel tracks. The first extends Theorem 4.2 to the planar entry problem, where the correspondence above is exact and Assumptions 2.2 and 2.3 are verified rather than assumed; the small question of whether the logarithm in $\varrho(\varepsilon)$ can be removed belongs to it. The second attacks objectives 1 and 2 together, since the threshold that replaces Assumption 2.4 and the amplitude that the moving section forces into the limit system are the same cut in the same family of oscillations. The third is objective 3, and it should be reached early enough that a negative result is still recoverable; objective 4 runs alongside it, needing only that the limit system be settled. If objective 3 resists a proof where frequencies cross, the retreat is a corridor on which the frequency bands are ordered, with the crossings at the edges of each pass left to the enclosure of the bubble, which integrates the coupled field and needs no separation; that weakens the residual without invalidating the decomposition.

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